

eVault: Complete Live Demonstration Script for Judges

Step-by-Step Presentation Guide: Stakeholder Workflows, RBAC, and Blockchain Anchoring

Step 0: The 30-Second Opening Pitch (Problem & Solution)

What to Say to the Judges:

*'Respected Judges, traditional court records suffer from two major flaws: vulnerabilities to database tampering and unauthorized data leakage. **eVault** solves this by combining **AES-256-GCM encryption**, **decentralized IPFS storage**, **Ethereum Sepolia blockchain anchoring**, and strict **Role-Based Access Control (RBAC)**. Today, I will demonstrate how 3 key stakeholders interact seamlessly with zero trust.'*

Step 1: Advocate (Lawyer) Flow — Case Creation & Document Filing

- 1. Connect & Sign In:** Connect MetaMask wallet and log in as Advocate (e.g. *Adv. Ramesh Sharma*, Bar: *MAH-10492-2020*). Explain that the wallet acts as the cryptographic signing pen.
- 2. Register New Case (Exclusive to Lawyer & Police):** Navigate to the **Cases** tab. Click **+ CREATE NEW CASE**. Enter Case ID *CASE-MH-501* (Title: *Sharma Industries vs. State of Maharashtra*, Assigned Judge: *Hon. Justice D. Y. Patil*). Click Register. Highlight to judges: *Only Lawyers and Police have authority to create legal cases.*
- 3. Encrypted Document Filing:** Go to the **Documents** tab. Select Case *CASE-MH-501* and attach a legal PDF. Click **ENCRYPT & SECURE ON BLOCKCHAIN**. Point out the live pipeline: *AES-256-GCM Encryption → IPFS Blob Pinning → Sepolia Metadata Registration → Audit Log.*

Step 2: Judicial Officer (Judge) Flow — Bench Orders & Judicial Signing

- 1. Switch to Judge Account:** Log out from Lawyer, and log in with the **Judicial Officer** role (e.g. *Hon. Justice D. Y. Patil*).
- 2. Demonstrate Separation of Powers:** Click the **Cases** tab. Point out the badge **'Case Creation: Advocates & Police Only'**. Explain that judges preside over matters and cannot initiate cases.
- 3. Issue On-Chain Court Order:** Go to the **Orders** tab (Bench Portal). Select the docket for *CASE-MH-501*. Enter the order text (e.g. *'Bail is granted upon furnishing a personal bond of Rs. 50,000'*) and click **SIGN & REGISTER JUDICIAL ORDER**. Show the live on-chain Sepolia multi-signature receipt.

Step 3: Citizen / Client Flow — Strict 3-Functionality Vault & Verification

- 1. Switch to Client Account:** Log out and log in with the **Citizen / Client** role.
- 2. Demonstrate Strict Permission Lockdown:** Look at the navigation ribbon—it is strictly locked to **My Vault & Upload** and **Audit**. Clients cannot browse arbitrary court cases and cannot fetch or download external documents.
- 3. Document Upload & Cryptographic Verification:** Upload an affidavit via **Upload Document**. Go to **My Vault** and click **VERIFY INTEGRITY**. Show the **Cryptographic Proof Report**: matches IPFS CID and SHA-256 hash directly with Sepolia to prove document authenticity.

Step 4: Live Transparency Audit Trail (The Closing Impact)

- 1. Open Audit Tab:** Walk the judges through the immutable audit ledger. Show every action performed during the demo: *USER_REGISTERED*, *CASE_REGISTERED_ON_ECOURTS*, *DOCUMENT_UPLOADED*, and *JUDICIAL_ORDER_SIGNED* with timestamps, hashes, and wallet addresses.

Closing Statement to Judges:

'As demonstrated, eVault delivers complete judicial integrity: physical identity is bound through Aadhaar e-KYC, records are tamper-proof on Ethereum, and all stakeholders operate within cryptographically enforced boundaries. Thank you!'

Top Judge Questions & Winning Answers:

• **Q: Why do we connect an Ethereum wallet?**

A: Email/Password authenticates the user into the portal; the Wallet provides the cryptographic private key required to digitally sign filings, register hashes on Sepolia, and ensure legal non-repudiation.

• **Q: Is sensitive case data stored publicly on blockchain?**

A: No. Full documents are encrypted with AES-256-GCM and stored off-chain on IPFS. Only zero-knowledge SHA-256 hashes and metadata are anchored on Ethereum.